Graph-based Programming Frameworks for Integrated Photonic Switching in Networking Applications

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1 Introduction

Motivation

Talk about limitations of electrical integration circuit and challenges in the post-moore's law era:

- Explosive growth in data transmission demand due to AI, ML..., Electronic processor reaches it's limitations, slow moore low, power, bandwidth, I/O bottle neck
- Requirements of new paradigm AI data center: energy efficiency, low latency, interconnect bandwidth, dynamic...

Talk about the solution of photonic processor for datacenter, optical interconnect:

- Photonic integrated circuits are solutions, high bandwidth, low latency, passive, low power consumption
- Optical interconnects are solutions for AI data transmission, especially optical circuit switch, talk about advantages, network reconfiguration

Talk about the state of the art of ocs applications in datacenter. These leading industry demonstrate that OCS become a strategic technology choice for overcoming performance and energy efficiency bottlenecks in AI data centers.

- google palomar, spine replacement, dynamic network
- nvidia paper, failure resilience

Talk about the importance of photonic hardware programming:

- Photonic integrated circuits they are programmable (PIP through MZI)
- Design and control complexity of the large-scale photonic processor

- Need a high level control layer to manage and configure photonic processor based products in datacenter
- Missing an application oriented programming abstractions for photonic processor design and controlling

Objectives

Research objectives:

- Provide an advanced, efficient, and universal programming methodology for programmable photonic processors in AI data centers
- Resolve how to efficiently map the datacenter applications to the photonic processor
- Address how to comprehensively design and simulate large scale OCS for datacenter
- Explore and develop an SDN of the layer 1 OCS for AI datacenter applications controller

Key contributions:

- Graph-based advanced programming methodologies for photonic applications
- Framework for Optical Circuit Switch Architecture Design
- SDN control system for Optical Circuit Switch

Structure of the thesis

Chapter 2: Theoretical and Programming Methodologies of Photonic Integrated Circuits

- talk about phase shifter / MMI / MZI
- circuits, feedforward / circular
- graphic theory, how to syntheses circuit based on graph
- sdn theory

Chapter 3: Optical Networking Applications Based on Programmable Integrated Photonic

- interconnect (result)
- beamsplitter (algorithm + result)
- switch (algorithm + result)

Chapter 4: optical circuit switch architectures for datacenter applications.

- type of topologies Benes/Dilated Benes/Piloss/Dilated banyan/Double Dilated Benes
- blocking study packets traffic / ocs traffic
- sax simulations (ILs / SNR)
- VPI link simulation with a STD SOA specification?

Chapter 5: software defined network for OCS

- custom yang model
- netconf
- gnmi

Chapter 6: discussions

- address large-scale ocs (losses / crossing) problem
- ocs datacenter use cases maturity (reconfigurable topologies challenging to implement)
- missing an AI network optimization control software
- sdn standardization process
- conclusions
- future work
  - OpenConfig
  - new architectures
  - multicore approaches